

EIS23
Initial Sizing & Longitudinal Control Design



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Abstract

An initial sizing was carried out for the conceptual design of Lelantus using a mixture of empirical and derived estimations. An estimated maximum takeoff weight of 524 kg and a wing area of 12.9 m^2 was sized, along with a sea level power rating of 44.2 kW for propulsion, which formed the initial design point. A set of longitudinal controllers for altitude and airspeed were also developed according to the requirements of NATO's STANAG 4671 and RPV Flying Qualities. The controller was verified and tested in both Simulink and X-Plane 10 flight simulator where it satisfied the majority of controller design goals.

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List of Symbols

AR	Aspect ratio
C	Power specific fuel consumption
C_D	3D drag coefficient
C_{D_0}	3D zero-lift drag coefficient
C_L	3D maximum lift coefficient
C_{L_L}	3D maximum lift coefficient in landing configuration
$C_{L_{TO}}$	3D maximum lift coefficient in takeoff configuration
D	Drag
D	Disturbance/gust model
E	Endurance
e	Oswald efficiency
h	Altitude
K_P	Proportional gain
K_I	Integral gain
K_D	Derivative gain
L	Lift
$(L/D)_{max}$	Maximum lift-to-drag ratio
N	Derivative filter coefficient
P	Propulsion power
q	Pitch rate
R	Range
S_L	Landing distance
S_{ref}	Wing reference area
S_{TO}	Takeoff distance
S_{wet}	Wetted surface area
U	Forwards velocity
V_i	Indicated airspeed
$V_{i_{minD}}$	Minimum drag indicated airspeed
V_{minD}	Minimum drag true airspeed
V_S	Stall airspeed
V_∞	True airspeed
$\bar{V}$	Non-dimensional airspeed
W	Downwards velocity
W_0	Maximum takeoff weight
W_e	Empty weight
W_f	Fuel weight
$W_{payload}$	Payload weight
α	Angle of attack
α	Weight fraction
β	Power lapse
γ	Climb gradient
δ_E	Elevator deflection
δ_{R1}	Left ruddervator deflection
δ_{R2}	Right ruddervator deflection
$\lambda_{0.25c}$	Quarter chord sweep
ρ	Air density
ρ_0	Sea level air density
η_p	Propeller efficiency
θ	Pitch
ω	Frequency
ω_D	Disturbance frequency

1 Introduction

Unmanned aerial vehicles (UAV) had become a critical industry technology, from agriculture to urban air logistics. Due to its unmanned nature, it is often cheaper to procure and operate, more capable in long hour operations within hazardous environment, compared to traditional manned aircraft. Recent conflicts have also highlighted the increased demand of more capable UAVs in a multitude of complex combat mission profiles [1].

Following market research, it was found that there is a market gap for long endurance, low observability surveillance drones with optional armaments at a weight class of less than a ton [2]. Therefore, an initial sizing must be conducted to design for the desired size category whilst achieving the target mission profile as defined in the executive summary [3]. The initial sizing formed the foundation work of the first design iteration. Subsequently, longitudinal controllers were also developed and are the primary focus of this report, ensuring that the defined performance and robustness goals were met as described in Section 3.

2 Initial Sizing

The initial weight sizing of the aircraft was conducted by analysing the mission segments using the Breguet's range and endurance equations, which required a number of empirical estimations. Namely, the max lift-to-drag ratio $(L/D)_{max}$, power specific fuel consumption, empty weight fraction W_e/W_0 and mission segment weight fractions. The UAV will be designed to carry a payload weight of 140 kg and carry out the specified mission profile. The final maximum takeoff weight (MTOW) estimation will be

$$W_0 = \frac{W_{payload}}{1 - (W_f/W_0) - (W_e/W_0)} \quad (1)$$

where fuel weight fraction W_f/W_0 and empty weight fraction W_e/W_0 will be derived. The weight sizing was followed by a sizing to constraint such that the aircraft is compliant with NATO's STANAG 4671 airworthiness requirements [4].

2.1 Empirical Values

As outlined in the executive summary [3], the baseline configuration was set as a flying wing body with an inverted V-tail attached via twin tail booms. The estimated max lift to drag ratio was primarily a function of the aircraft's wetted area ratio S_{wet}/S_{ref} [5] using Equation 2. A pure flying wing has a wetted ratio of slightly above 2 as its surface area is the double of its wing planform area. A well designed twin boom tail with a long moment arm and a low tail and boom area will have little contribution to the overall surface area. However, a more conservative estimate of wetted area ratio of 3 was taken.

$$\left(\frac{L}{D}\right)_{max} = K_{LD} \sqrt{\frac{AR}{S_{wet}/S_{ref}}} \quad (2)$$

The empirical modifier K_{LD} is dependent on the type of aircraft. For the chosen aspect ratio of 10, K_{LD} was estimated as 13 [5] which resulted in an estimated maximum lift-to-drag as 23.7.

The power specific fuel consumption was predicted as 285 g/kW/hr based on the powertrain selection [6]. Due to the hybrid nature of the powertrain system where energy losses will occur between engine and generator, generator and motors, a 95% efficiency factor was applied, resulting in a power specific fuel consumption of 300 g/kW/hr.

Certain mission segments' weight fractions were taken empirically based on Roskam's suggestions [7]. Weight fractions of 0.98 and 0.99 for climb and descend respectively were estimated, taken from conventional airliners that climb to and descend from high altitudes of 35000 ft. Therefore, the actual weight fraction consumption was linearly interpolated for the altitude change of the specified mission profile. The weight fraction for taxi and takeoff was estimated using a higher estimate at 0.99. Since operationally, the drone will be pushed onto the road for startup and takeoff without the need of a lengthy taxi phase across an airport. Lastly, landing was estimated to be 0.995.

2.2 Target Design Parameters

In addition to empirical predictions, several target design values must also be set in order to size the aircraft. For example, the target clean maximum lift coefficient $C_{L_{max}}$ was estimated as 1.35 which was obtained from a predicted maximum 2D aerofoil lift of 1.6 with a 0.9 modifier for 3D wing effects. Another modifier of $\cos(\Lambda_{0.25c})$ was also applied due to sweep. However, since only the leading edge sweep was set at 20° and the wing planform consists of multiple cranks, the quarter chord sweep $\Lambda_{0.25c}$ was dependent on the final planform design and was also constrained to be less than 20° . Thus, a quarter chord sweep was assumed at the maximum 20° for a conservative prediction. In landing configuration, the maximum lift coefficient $C_{L_{maxL}}$ was predicted to be 1.51 by assuming the use of plain flaps which resulted in an increase of 2D max lift coefficient of 0.9 [8], with a flapped area ratio of 20%. The flapped area ratio was also heavily dependent on crank locations. Thus, a low value was assumed for a conservative estimate. Takeoff maximum lift coefficient $C_{L_{maxTO}}$ was assumed to be 1.46 by assuming 70% total lift coefficient increase from partial deflection of flaps. Last but not least, a target Oswald efficiency e of 0.8 was chosen which resulted in a zero lift drag coefficient C_{D_0} of 0.011.

2.3 Weight Fractions

2.3.1 Fuel Weight Fraction

Using Breguet's range and endurance equation for power generating aircraft, the cruise and loiter weight fractions were given by

$$\frac{W_i}{W_{i-1}} = \exp\left(\frac{-RC}{\eta_p(L/D)}\right) \quad (3) \quad \frac{W_i}{W_{i-1}} = \exp\left(\frac{-ECV_\infty}{\eta_p(L/D)}\right) \quad (4)$$

respectively [5] where R is the specified range, C is the power specific fuel consumption, and η_p is the propeller efficiency which was assumed to be 0.8. Furthermore, E is the specified endurance and V_∞ is the true air speed. All values are in S.I. units. The lift to drag ratio L/D for steady level flight was given by

$$\frac{L}{D} = \left(\frac{L}{D}\right)_{max} \frac{2\bar{V}^2}{\bar{V}^4 + 1} \quad (5) \quad \bar{V} = \frac{V_\infty}{V_{minD}} = \frac{V_i}{V_{i_{minD}}} \quad (6)$$

where $\bar{V}$ is the non-dimensional speed given as a fraction of the minimum drag speed V_{minD} . A 2% fuel reserve was implemented as a safety margin. Consequently, it resulted in a fuel weight fraction of 23%.

2.3.2 Empty Weight Fraction

Due to the nicheness of the baseline configuration, literature does not provide an adequate empirical prediction of the empty weight fraction. Therefore, a custom weight fraction database of long endurance combat UAVs was developed and built as shown in Table 5. A curve of form

$$\frac{W_e}{W_0} = AW_0^C \quad (7)$$

was fitted where the empirical values A and C was found at 0.956 and -0.103 respectively. The coefficient of determination R^2 value was 0.1, showing a poor data fitting. Due to the experimental nature of the baseline configuration, and the non-standardised design doctrines of military UAVs, empirical weight predictions were not accurate. Therefore, the initial size can only serve as an initial design point. In order to finalise the final MTOW, a minimum drag speed in equation 6 must be defined. Figure 1 shows the effects of varying minimum drag speed (which is chosen as the cruise and loiter speeds). A slower cruise speed is more fuel efficient but requires a larger wing. A compromise speed of 35 m/s was chosen such that the wingspan still satisfied field width clearance of 16 m, whilst providing sufficient wing area for packaging. The resulting empty weight fraction and final MTOW estimate is outlined in Table 1.

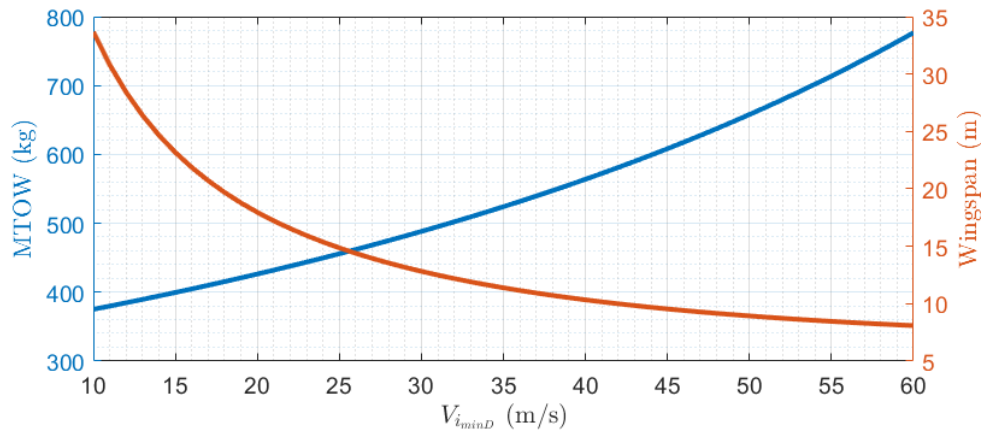


Figure 1: Effect of minimum drag speed

Table 1: Aircraft Weight and Weight Fractions

Weight Type	Weight (kg)	Weight Fraction
MTOW	524	1.00
Payload	140	0.27
Fuel	122	0.23
Empty	262	0.50

2.4 Sizing to Constraints

The primary constraints as stated in STANAG 4671 were that takeoff climbout must achieve a minimum gradient of 5% at climbout speed of $1.3 V_S$, and landing must clear a 50 ft vertical obstacle [4]. Furthermore, from MIL-C-5011, it must achieve 500 fpm climb rate at the combat altitude [9]. Due to the experimental nature of the design space, there were very few constraining specifications. Other self-imposed constraints based on operational requirements [2] include

- Loiter and cruise flight phases
- 45° level banked turn at cruise (equivalent 1.4 G load factor) to ensure sufficiently tight turning radius of less than 200 m for weapons employment
- Takeoff and landing on a 400 m paved field for operations out of improvised road bases
- Sea level climb rate of 1000 fpm to ensure sufficiently short time to altitude
- Absolute ceiling at 35000 ft

Figure 2 shows the final constraint diagram which was constructed using equations 16 to 22. The wing loading was constrained to 397 N/m^2 by the chosen minimum drag speed and assumed aerodynamic properties, which resulted in a wing area of 12.9 m^2 and wingspan of 11.4 m . Taking the lowest possible power to weight ratio of 8.6, it required a sea level power rating of 44.2 kW for propulsion.

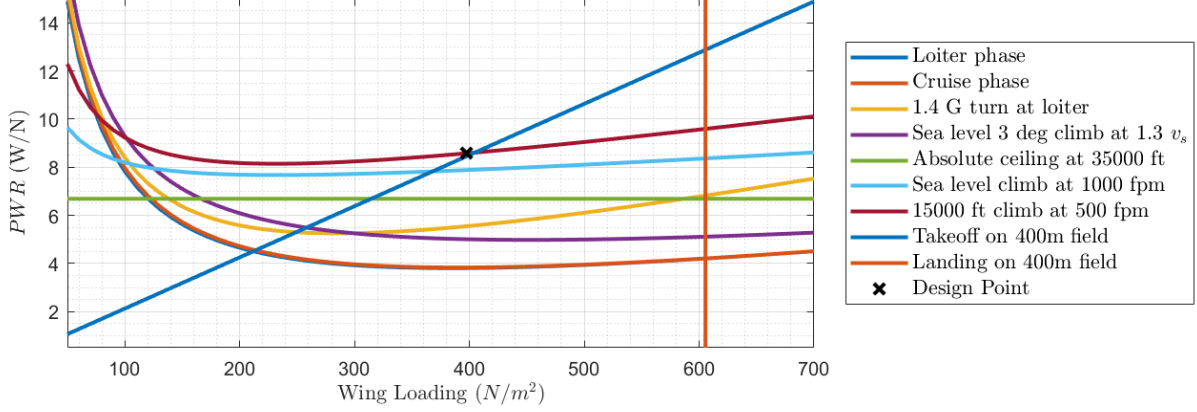


Figure 2: Constraint diagram

3 Longitudinal Control

As part of the development work for longitudinal controllers, an altitude controller and a throttle controller were developed which output direct commands to two control input channels: the throttle u_T and elevator u_{δ_E} . The controllers were developed with the primary goals of step reference tracking, disturbance rejection and robustness. Due to the inverted V-tail configuration, the elevator inputs were superimposed onto the rudder inputs u_{δ_R} by summation, which was fed to the left δ_{R1} and right ruddervators δ_{R2} with a one to one mixing ratio as modelled in the flight simulator [10]. Whereas the throttle input commands thrust to both electric motors.

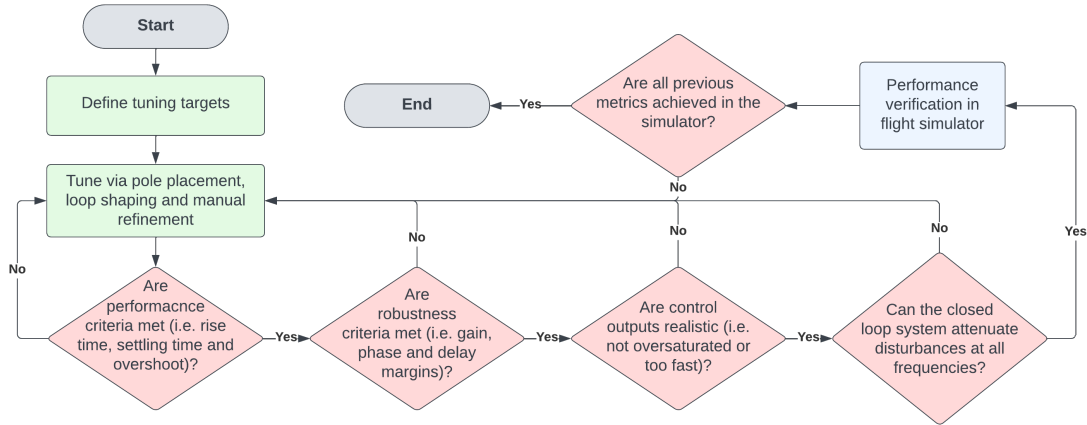


Figure 3: Tuning Methodology

The scope of the longitudinal controller development will be limited to the cruise condition at the start of the loiter phase which is clean configuration at 66 kn IAS and 15000 ft , with 87% fuel load and full armaments. The linearised state space dynamic model of the aircraft at the aforementioned conditions was represented by

$$\begin{bmatrix} \dot{U} \\ \dot{W} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -0.024 & 0.139 & 0.033 & -9.811 \\ -0.452 & -2.123 & 40.902 & 0 \\ 0.028 & -0.296 & -7.823 & 0 \\ 0 & 0 & 1.000 & 0 \end{bmatrix} \begin{bmatrix} U \\ W \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} 0.914 & 0.070 & 0.070 \\ 0 & 1.589 & 1.589 \\ 0 & 9.621 & 9.621 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u_T \\ u_{R1} \\ u_{R2} \end{bmatrix} \quad (8)$$

as derived by Surkov et al [11] [12] [13].

Parallel form continuous time PID with derivative filter was selected for the primary control method due to its ability to manipulate the closed loop system behaviour via pole placement. Furthermore, the derivative filter coefficient, N , enabled the filtering of high frequency noise to prevent its amplification by the derivative gain. The controller took the form

$$K(s) = K_P + K_I \frac{1}{s} + K_D \frac{N}{1 + N \frac{1}{s}} = \frac{s^2(K_P + K_D N) + s(K_I + K_P N) + K_I N}{s(s + N)} \quad (9)$$

which allowed the placement of two additional zeros and a pole in order to achieve the desired closed loop behaviour. The tuning methodology is outlined in Figure 3 with a typical tuning workspace shown in Figure 22.

3.1 Controller Structure

Due to the complex dynamics of an aircraft's longitudinal motions, controlling the aircraft with a single loop closure can often result in undesirable oscillatory behaviour which may even lead to instability [14]. For example, the first iteration of climb gradient controller had a single closure loop where a single PID controller actuates the elevator. Due to the oscillatory coupling between angle of attack and pitch rate (and therefore pitch angle), the closed loop behaviour was often oscillatory or had an overshoot as shown in Figure 4. It also had a finite gain margin and a marginal phase margin. Most importantly, the closed loop behaviour amplified disturbances at its resonant frequency as shown in Figure 5. When verified in the simulator, the closed loop behaviour worsened and became oscillatory due to plant model mismatch as well as transport delays that exist in the simulator of the order of magnitude of 0.01 s, depending on load of the local network.

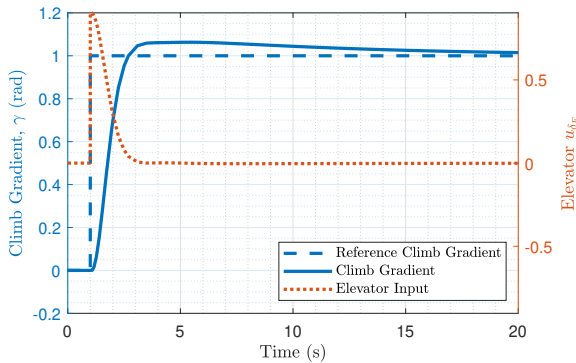


Figure 4: Single loop climb gradient unit step tracking

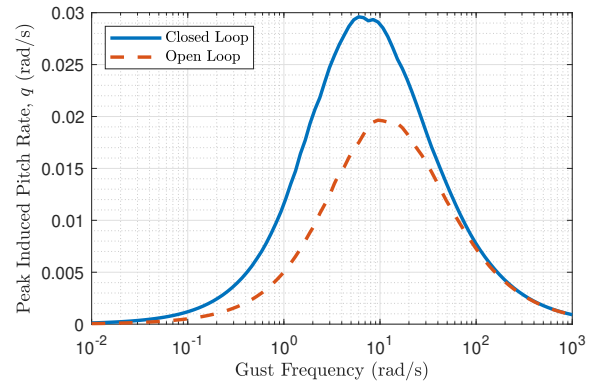


Figure 5: Single loop climb gradient controller across frequencies

Therefore, the final altitude controller consisted of a triple cascaded loop layer, each with a PID controller as shown in Figure 6. A well tuned cascaded PID controller can reject high frequency noise as the most inside loop has the highest bandwidth frequency which can quickly respond

to any perturbation with minimal effects on the outer loop's tracking [15]. The triple loops consisted of an outer altitude loop, middle climb gradient loop, and inner pitch rate loop. The controllers were tuned using successive loop closure by tuning from the inside-out. The inner loops were assumed as unity gain when tuning the next outer loop [15]. The throttle controller consisted of a single loop PID controller. Although velocity was coupled with the aircraft's pitch angle by the Phugoid mode, its natural frequency, at 0.26 rad/s, was far lower than that of short period pitching oscillation (SPPO) associated with pitch control, at 5.36 rad/s. Therefore, a single loop PID was deemed sufficient as a high frequency pitch controller could ensure stable tracking in pitch.

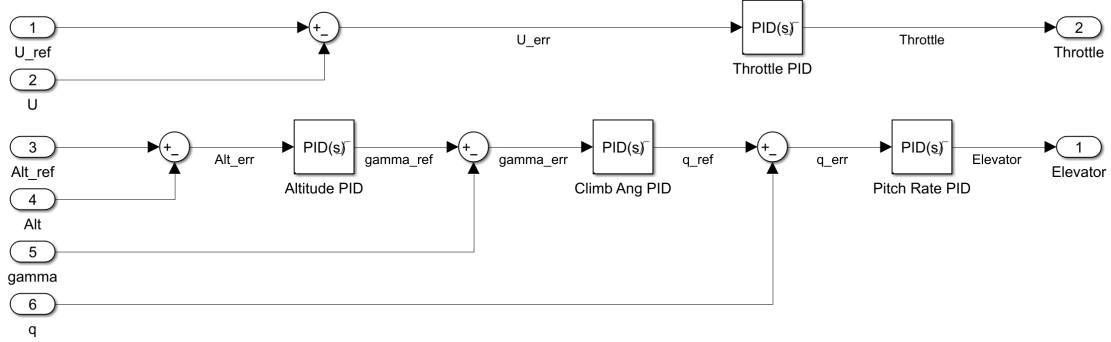


Figure 6: Structure of the longitudinal controllers

3.2 Tuning Goals

To ensure compliance with STANAG 4671 [4] and RPV (Remotely Piloted Vehicle) Flying Qualities [16], the following criteria must be met

1. Altitude hold must have a minimum control accuracy of ± 30 ft
2. Pitch response must be "smooth and rapid" with less than 20% overshoot
3. Airspeed hold must be capable of maintaining IAS within ± 5 knots
4. Flight path (climb angle) stability for change in airspeed must be lower than $0.06^\circ/\text{kn}$
5. Gain and phase margins must achieve 6 and 45° respectively
6. Must be able to endure gust magnitudes of 15.2 m/s

Rise time and settling time were both self-defined as shown in Table 2. The rise time was defined as the time taken for the response to reach 90% and settling time was defined as the time taken for the response to settle around $\pm 2\%$ of the setpoint. The rise times were selected such that the bandwidth frequency between different loops is separated by a factor of 5 to 10 such that individual controllers do not interact with each other [15]. A lenient rise time was defined for throttle controller for flight path stability. An aggressive throttle controller would have an adverse effect on flight path stability.

For a combat reconnaissance platform, a higher emphasis was placed on stability and smoothness rather than rapid response capabilities in order to maintain a stable observation over the target area, and to employ and provide guidance to precision guided munition. Therefore, a 5% maximum overshoot target was set to all controllers in order to minimise oscillations. Ideally, the closed loop step tracking behaviour should be over-damped such that no oscillatory motion would occur. In addition to the stipulated gain and phase margins, a transport delay margin of 0.1 s was also set. Unlike phase delays which affect frequencies by a phase offset, transport delay due to latency across the network shifts all frequencies equally within the time domain [17]. Therefore, special consideration must be given to it for robustness design criteria.

Lastly, the controller must be able to reject disturbances across a broad range of frequencies ω_D spanning 10^{-2} rad/s to 10^3 rad/s, such that specific frequencies of wind gust does not destabilise the closed loop response. These frequency ranges were chosen such that it encompasses the natural frequencies of Phugoid and SPPO, as it was suspected that the destabilising resonant frequencies will be close to the natural frequencies of the open loop dynamics. The peak amplitude of the closed loop system should not be higher than that of the open loop peak. A smooth single frequency perturbation, $D(t)$ of the form

$$D(t) = \begin{cases} \frac{1}{2}(1 - \cos(\omega_D t)) & \text{if } 0 \leq t \leq \frac{2\pi}{\omega_D} \\ 0 & \text{if } t > \frac{2\pi}{\omega_D} \end{cases} \quad (10)$$

was used to excite the closed loop controllers at a range of frequencies.

Table 2: Step reference tracking performance goals of various controllers

Controller	Rise Time (s)	Settling Time (s)	Overshoot (%)
Throttle	5.0	8.0	5.0
Pitch Rate	0.5	2.0	5.0
Climb Gradient	3.0	6.0	5.0
Altitude	15.0	30.0	5.0

3.3 Individual Controller Tuning

3.3.1 Pitch Rate Controller

The pitch rate controller of the most inner loop was tuned using a modified state space matrix with only the relevant states of SPPO, downwards velocity W and pitch rate q , such that

$$\begin{bmatrix} \dot{W} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} -2.123 & 40.902 \\ -0.296 & -7.823 \end{bmatrix} \begin{bmatrix} W \\ q \end{bmatrix} + \begin{bmatrix} 1.589 & 1.589 \\ 9.621 & 9.621 \end{bmatrix} \begin{bmatrix} u_{R1} \\ u_{R2} \end{bmatrix} \quad (11)$$

where the pitch rate controller outputs the required elevator input u_{δ_E} which was then superimposed onto both ruddervator inputs, $u_{\delta_{R1}}$ and $u_{\delta_{R2}}$, using the error in pitch rate. The controller was first tuned on a unit step reference track without saturation effects as shown in Figure 7. The controller satisfied all targets stated in section 3.2 as shown in Table 4; its gains and performance metrics are summarised in Table 3. Subsequently, a realistic pitch rate of $30^\circ/\text{s}$, which was also set as the pitch rate limit, was used to test for saturation in the output response. The output was not saturated. The ruddervator servos were sized to be able to deflect fully under 0.4 s [18] at maneuvering speeds, which is higher than the cruise speed.

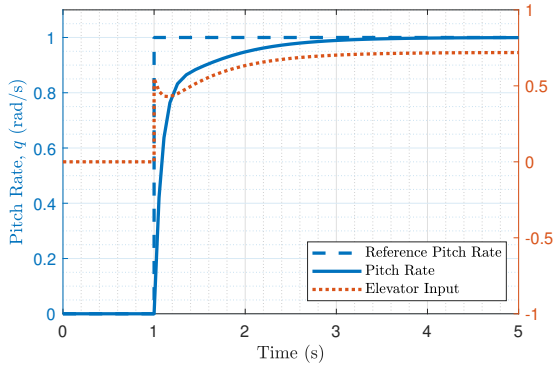


Figure 7: Pitch rate controller unit step tracking

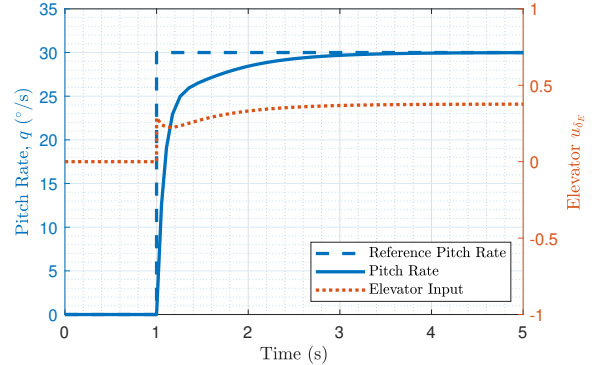


Figure 8: Pitch rate controller step tracking at $30^\circ/\text{s}$

The controller's ability to reject disturbance was also analysed by exciting the W state of the system using the perturbation model in Equation 10. As shown in Figure 9, the closed loop behaviour could effectively attenuate perturbation at the range of frequencies considered. The closed loop peak perturbed state change was only 48.4% of the open loop's.

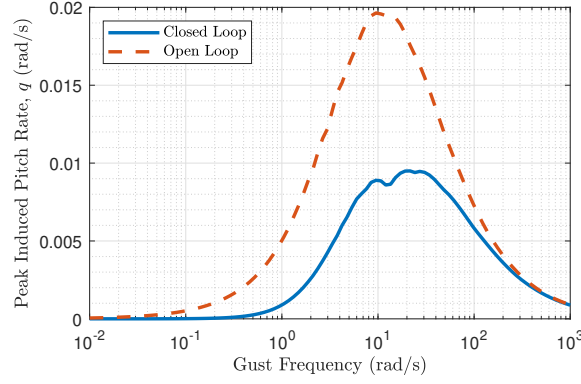


Figure 9: Pitch rate open and closed loop response to perturbation

3.3.2 Climb Gradient Controller

The climb gradient controller was tuned by assuming unity gain on the pitch rate controller. Therefore, any reference pitch rate commanded by the climb gradient controller becomes the pitch rate of the aircraft. By further modifying the state space matrix,

$$\dot{W} = -2.123W + 40.902q_{ref} \quad (12)$$

such that the reference pitch rate q_{ref} became an input to a state space with a single state, W . The climb gradient was then

$$\gamma = \theta - \alpha \approx \frac{q_{ref}}{s} - \frac{W}{U_e} \quad (13)$$

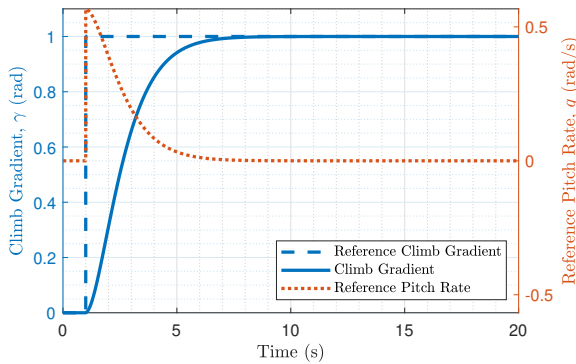


Figure 10: Climb gradient controller unit step tracking

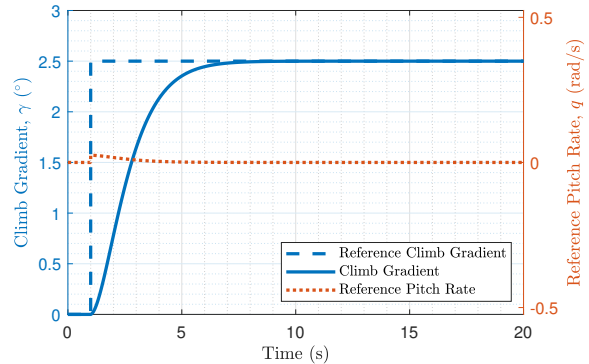


Figure 11: Climb gradient controller reference step tracking at 2.5°

where U_e is the equilibrium velocity at which the dynamic modes were linearised about and s is the Laplace variable. The unit step tracking performance, without saturation, is shown in Figure 10. It achieved all stipulated goals in section 3.2 whilst achieving zero overshoot nor oscillations. Figure 11 shows the closed loop behaviour using a realistic climb gradient step tracking of 2.5° . The output reference pitch rate was satisfactory. It did not oversaturate the

imposed pitch rate limit of $30^\circ/\text{s}$ (0.52 rad/s). Disturbance rejection was not analysed for the climb controller as it was assumed the most inner loop will be able to remove the perturbation before reaching the outside loop. Instead, disturbance rejection will be analysed for the full cascaded controller.

3.3.3 Altitude Controller

The altitude controller was tuned by assuming unity gain on both the climb gradient controller and throttle controller. The altitude, h , was given by

$$h = \frac{U_e \sin(\gamma_{ref})}{s} \quad (14)$$

where the reference climb gradient γ_{ref} was the output of the altitude controller and it was assumed that the throttle controller could maintain the airspeed at equilibrium velocity U_e . All stipulated tuning targets were met as shown in Figure 12. However, unlike the previous controllers, the altitude controller saturated easily as shown in Figure 13. The 2.5° climb gradient limitation was imposed due to the predicted decayed climb performance at cruise compared to sea level [13]. Despite the saturation, the response had no overshoot nor oscillations due to the implementation of anti-integral windup algorithm.

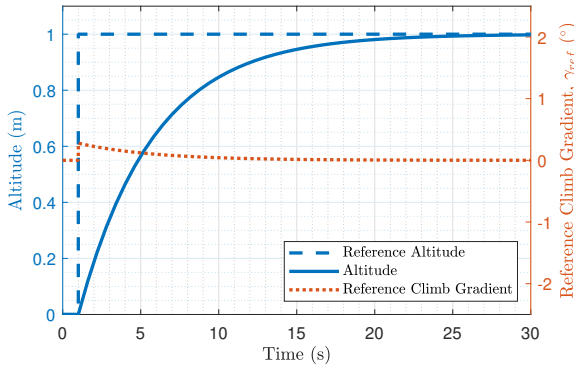


Figure 12: Altitude controller unit step tracking

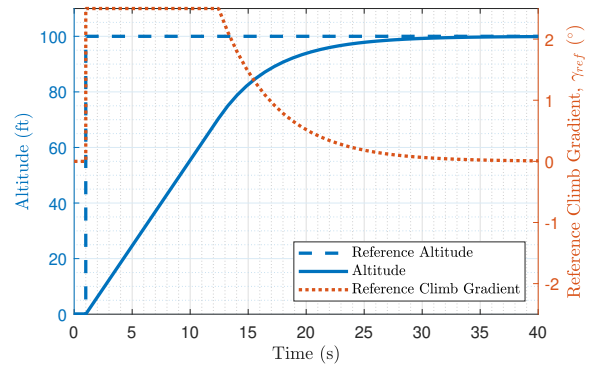


Figure 13: Altitude controller reference step tracking of 100 ft

3.3.4 Throttle Controller

The throttle controller was tuned by isolating the forwards velocity state U such that

$$\dot{U} = -0.024U + 0.914u_T \quad (15)$$

Such simplification eased the tuning process. However, the implication was that pitch must not vary significantly such that the aircraft is flying pitch level and gravity does not influence its airspeed. Furthermore, due to the low natural frequency of Phugoid, it was also assumed that the pitch rate controller, with a much higher bandwidth frequency, could remove any perturbation in pitch such that it can be decoupled from velocity. With these assumptions, the throttle controller can be tuned to satisfy all goals set in section 3.2 as shown in Figure 14. Based on the cruise thrust setting of 46% [13], this placed a saturation limit on the throttle input u_T of -46% and 54% within the linearised state space such that the propulsion system does not exceed its max continuous thrust nor produce reverse thrust. As seen in Figure 15, the throttle saturated easily for a realistic increase in airspeed of 10 kn. However, due to the implementation of anti-integral windup, it did not impact performance. The controller also

excels in rejecting disturbances across the selected frequencies as seen in Figure 16, making it extremely robust in gusty conditions.

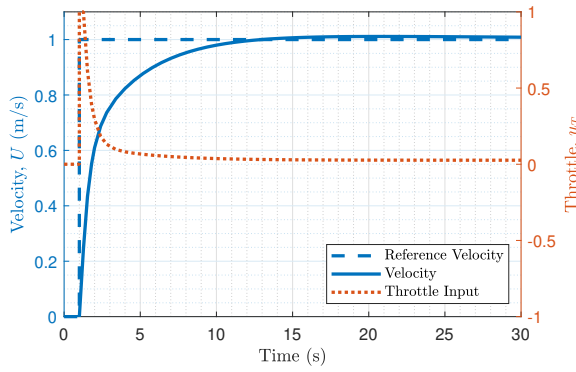


Figure 14: Throttle controller unit step tracking

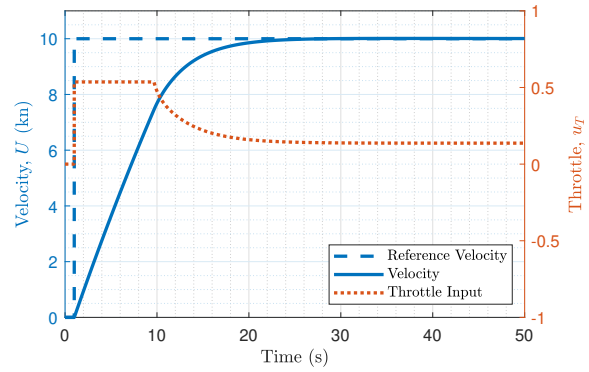


Figure 15: Throttle controller reference step tracking of 10 kn

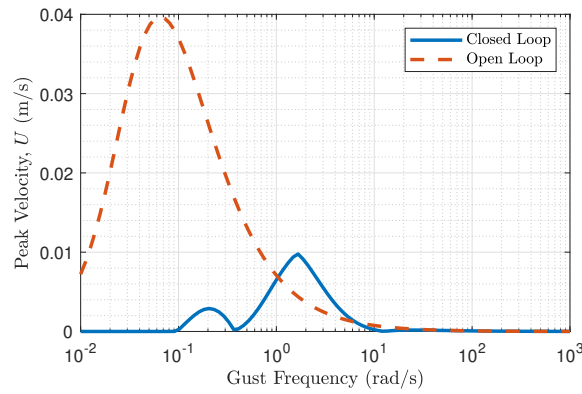


Figure 16: Airspeed open and closed loop response to perturbation

3.3.5 Individual Controller Gains and Performance

A summary of all controller gains are shown in Table 3 with their corresponding performance and robustness metrics shown in Table 4.

Table 3: Gains of all longitudinal controllers

Controller	K_P	Unit	K_I	Unit	K_D	Unit	Derivative Coefficient, N
Throttle	0.680	/ (m/s)	0.030	/m	0.880	/ (m/s ²)	1.00
Pitch Rate	0.746	/ (rad/s)	5.219	/rad	0.745	/ (rad/s ²)	0.10
Climb Gradient	0.559	rad/s/°	5.60e-8	rad/s ² /°	11.25	rad/°	6.95e-4
Altitude	0.199	°/m	2.00e-8	°/(ms)	465.72	°/(m/s)	6.54e-6

Table 4: Performance and robustness of all controllers

Controller	Rise Time (s)	Settling Time (s)	Overshoot (%)	Gain Margin	Phase Margin (°)	Delay Margin (s)
Throttle	4.24	7.91	1.82	∞	110	1.69
Pitch Rate	0.25	1.26	0.00	∞	96	0.11
Climb Gradient	2.97	5.08	0.00	∞	76	2.42
Altitude	14.5	25.7	0.00	∞	90	10.3

3.4 Integrated Longitudinal Controller Verification

The final integrated longitudinal controller as illustrated in Figure 6 was first tested and verified within a custom built Simulink framework shown in Figure 23, which utilised the full state space in equation 8. Figure 17 and Figure 18 shows the tracking of a step altitude increase and a step velocity increase respectively. The closed loop behaviour of altitude tracking and velocity tracking of each individual controller showed a close match with their predicted performance from section 3.3 when tracking their respective states. However, the coupling between pitch control and throttle control still impacted performance. In Figure 18, the increase in airspeed resulted in a $0.10^\circ/\text{kn}$ increase in climb gradient. Therefore, falling short of the stipulated $0.06^\circ/\text{kn}$ flight path stability to achieve level 1 performance, but still achieving level 2 with a stipulated flight path stability of $0.15^\circ/\text{kn}$ [16]. Similarly, changes in altitude resulted in transient behaviour of airspeed, albeit quite low at ± 1 kn as shown in Figure 17.

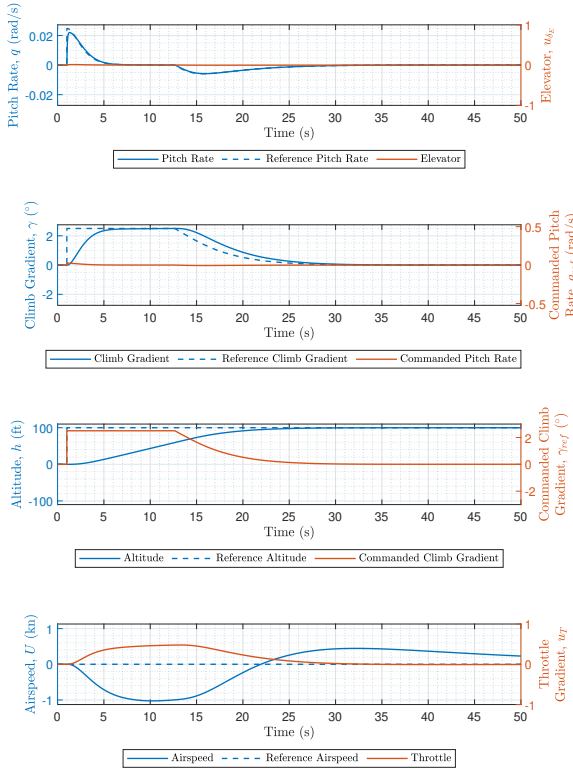


Figure 17: Integrated longitudinal controller in 100 ft climb

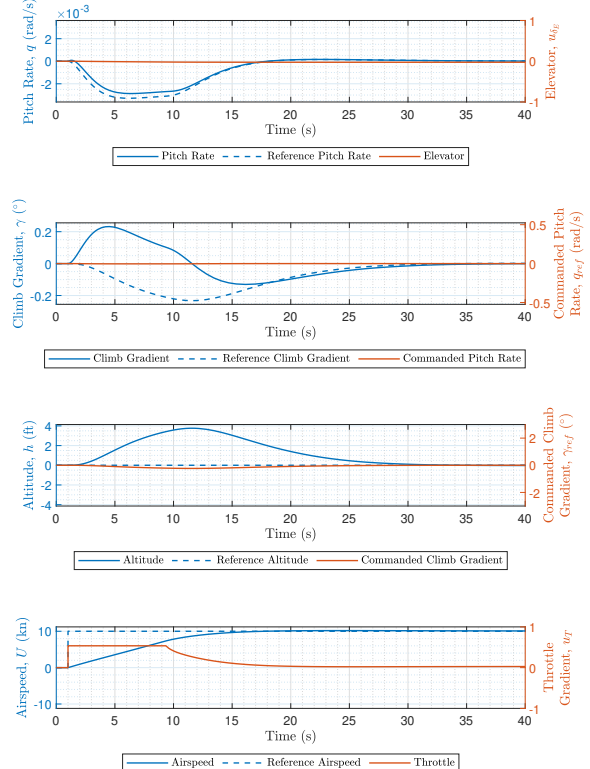


Figure 18: Integrated longitudinal controller in 10 kn airspeed increase

Figure 19 and Figure 20 shows the peak amplitudes induced by different frequencies of gust. The controller showed good rejection against perturbation in airspeed, with a peak reduction of 82%. However, the peak reduction of perturbation in downwards velocity was 39%. Whilst

the performance was still satisfactory such that it did not worsen the performance compared to the open loop, it shifted the resonant frequency to 2.7 rad/s. This was however expected, as it matched closely of that shown in the pitch rate controller in Figure 9.

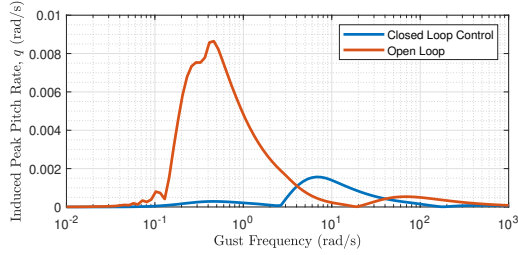


Figure 19: Disturbance rejection of the integrated longitudinal controller in forwards velocity U

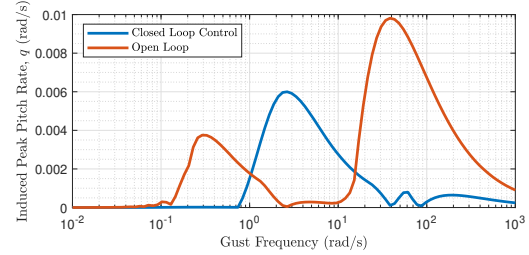


Figure 20: Disturbance rejection of the integrated longitudinal controller in downwards velocity W

The integrated longitudinal controller was also verified within the flight simulator at its cruising condition as shown in Figure 24 to 27.

4 Discussion and Future Works

The developed longitudinal controller met all goals set in section 3.2, barring flight path stability. Although the simplifications made when tuning individual controllers simplified the tuning process, it masked a number of couplings between the aircraft's dynamic states. As discussed in section 3.3.4, the throttle controller did not account for the effect of pitch on the aircraft. In reality, high pitch angles has a major impact on an aircraft's forces along its longitudinal axis due to the rotation of its weight force vector relative to the stability axis. In future works, another feedback controller should be introduced which takes in the pitch state of the aircraft and outputs a throttle command to counter pitch effects. Thus, mitigating the impact of pitch on airspeed tracking. Another controller type that could be developed is a feedforward controller. For example, the pitch rate controller could benefit greatly from a feedforward controller as the steady state pitch rate is directly dependent on the elevator deflection. Having two separate controllers increase performance by improving reference tracking and disturbance rejection, by separately tuning the feedforward and feedback controllers to track reference signals and reject noise/disturbance respectively.

In addition to couplings within the longitudinal states, there were also coupling with the lateral states that could only be observed via flight testing in the simulator. For example, the tracking performance of climb gradient was degraded when the aircraft was placed in a banked turn, as the roll controller was more aggressive than the climb gradient controller. A solution would be for the climb gradient controller to output an inertia axis reference pitch rate instead of a body axis reference pitch rate. The inertia axis pitch rate could then be transformed into body axis pitch rate which could then be effectively tracked by the pitch rate controller. Last but not least, controller performance was degraded when not flying at the pre-determined cruising conditions. For example, oscillatory pitching behaviour was observed in a dive at 200 kn in the simulator. At higher speeds, the control surface generated a higher force proportional to velocity squared. On the other hand, different flight phases have different control requirements. For example, max pitch rate should not be limited to $30^\circ/\text{s}$ to allow for landing flare before touchdown. Therefore, gain scheduling would be required to achieve desired flight characteristics between various flight regimes and configurations.

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A Empty Weight Fraction Estimation

A custom database of MTOW and empty weight fractions of various long endurance UAVs were built in order to estimate the empty weight fraction in equation 1. A curve which takes the form of equation 7 was fitted from the database shown in Table 5. Both the data fitting and MTOW prediction are visualised in Figure 21.

Table 5: Database of Long Endurance UAV

UAV Make & Model	MTOW (kg)	Empty Weight Fraction
Northrop Grumman Firebird	2300	0.51
DRDO Rustom 1	1800	0.40
Baykar Bayraktar TB2	700	0.60
Kronshtadt Orion	1150	0.43
Northrop Grumman MQ-4C Triton	14630	0.46
Vestel Karayel	560	0.85
Boeing Insitu RQ-21 Blackjack	61	0.61
Boeing X-45	10000	0.32
Denel Dynamics Bateleur	1000	0.20
Northrup Grumman X-47B	20220	0.31
General Atomics MQ-9 Reaper	4760	0.26
Kratos XQ-58 Valkyrie (XQ-222)	2722	0.42
General Atomics Altair	3175	0.46
Boeing MQ-28 Ghost Bat	3000	0.50
China Defence Rainbow CH-4	4500	0.36
Chengdu Wing Loong-10 Shadow	3200	0.72

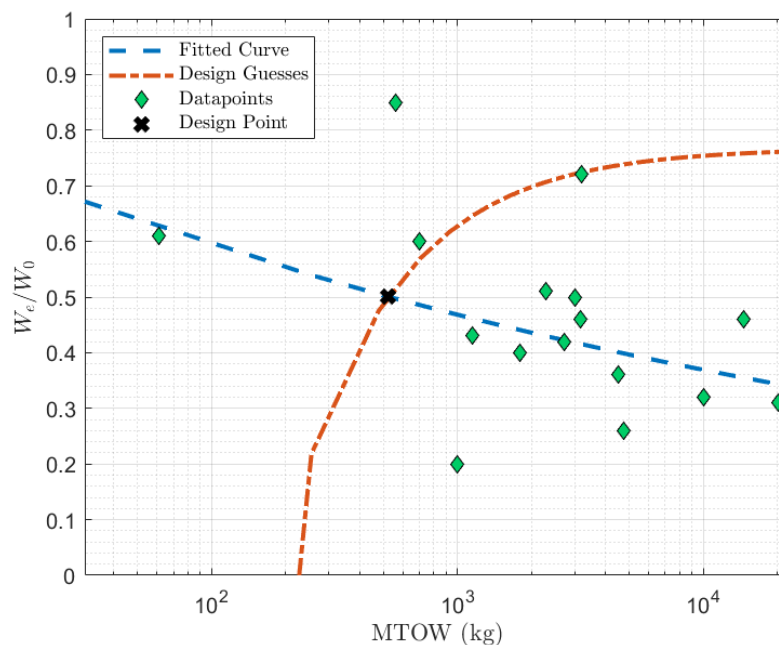


Figure 21: Empirical Estimation of Empty Weight Fraction

B Constraint Diagram Equations

The point performance constraints were calculated using specific energy equation [19] adapted for power producing aircraft such that the sea level power to weight ratio at MTOW is

$$\left(\frac{P}{W}\right)_0 = \frac{1}{\eta_p} \frac{\alpha}{\beta} \left[\frac{dh}{dt} + \frac{V_\infty}{g} \frac{dV_\infty}{dt} + \frac{\frac{1}{2}\rho V_\infty^3 C_{D0}}{\alpha W_0/S_{ref}} + \frac{\alpha n^2 W_0/S_{ref}}{\frac{1}{2}\rho V_\infty \pi A Re} \right] \quad (16)$$

where dh/dt and V_∞/dt are climb rate and acceleration respectively. The weight fraction α and power lapse β are given by

$$\alpha = \frac{W}{W_0} \quad (17) \quad \beta = \left(\frac{\rho}{\rho_0}\right)^{1.1} \quad (18)$$

with the power lapse of a non-supercharged piston engine assumed to be dependent on the ratio of air density [20]. The absolute ceiling constraint assumed the aircraft flew at its maximum lift-to-drag ratio such that

$$\left(\frac{P}{W}\right)_0 = \frac{\alpha V_\infty}{\beta \eta_p (L/D)_{max}} \quad (19)$$

The takeoff distance S_{TO} constraint was sized empirically using Roskam [7] where

$$S_{TO} = 8.134 TOP + 0.0149 TOP^2 \quad (20) \quad TOP = (W/S)_0 (W/P)_0 / C_{L_{max_{TO}}} \quad (21)$$

which is a function of the takeoff parameter TOP in units of $\text{lbs}^2/\text{ft}^2\text{hp}$. The landing distance S_L constraint was also sized empirically using method suggested by Roskam where

$$S_L = 0.5136 V_{S_L}^2 \quad (22)$$

which is a function of stall speed at landing V_{S_L} in knots. A field length safety margin of 67% and 50% was implemented for takeoff and landing respectively as suggested by Roskam for FAR-25 compliant aircraft to account for civilian pilot errors [7]. Despite better training of military pilots, such a high safety margin was used as improvised road bases will likely lack the infrastructure required for automated landings. GPS guided approaches are classified as non-precision instrument approach [21] and are prone to GNSS jamming. Thus, launch and recovery phases will likely be flown by a pilot remotely where latency may result in degraded control characteristics.

C Controller Tuning Workspace

A typical controller tuning workspace is shown in Figure 22 using Simulink's Control Systems Toolbox [22]. The initial tuning was completed by manipulating the zeros and poles of the PID controller shown in Equation 9 to position the root locus curve over the desired root positions. If necessary to achieve robustness criteria, the Nyquist and bode plots can be manipulated to shape the loop. Lastly, the exact gains could be manually refined. For dynamic modes with oscillatory coupling of states which proved to be difficult to tune, a cost minimisation method was employed to obtain the PID gains that would match a reference closed loop step tracking as closely as possible. Multiple controller gains could also be overlaid as shown by the green line in Figure 22 in order to compare and select the most appropriate controller design.

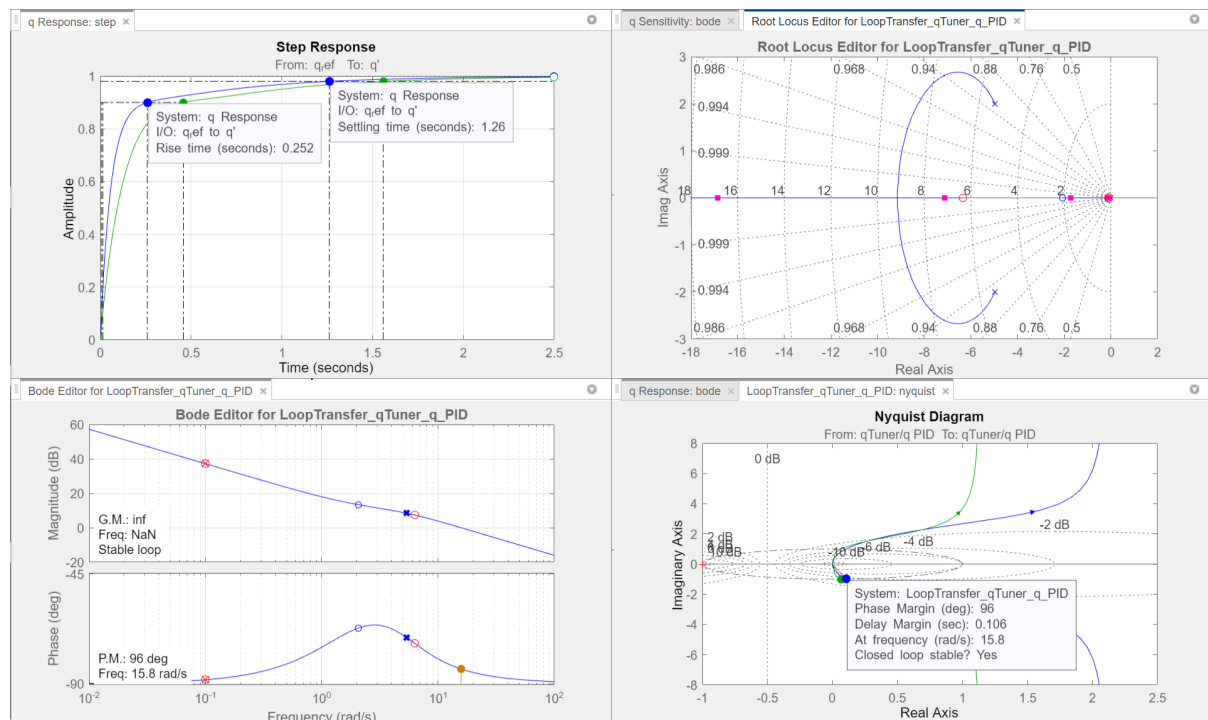


Figure 22: Typical controller tuning workspace

D Simulink Verification Framework

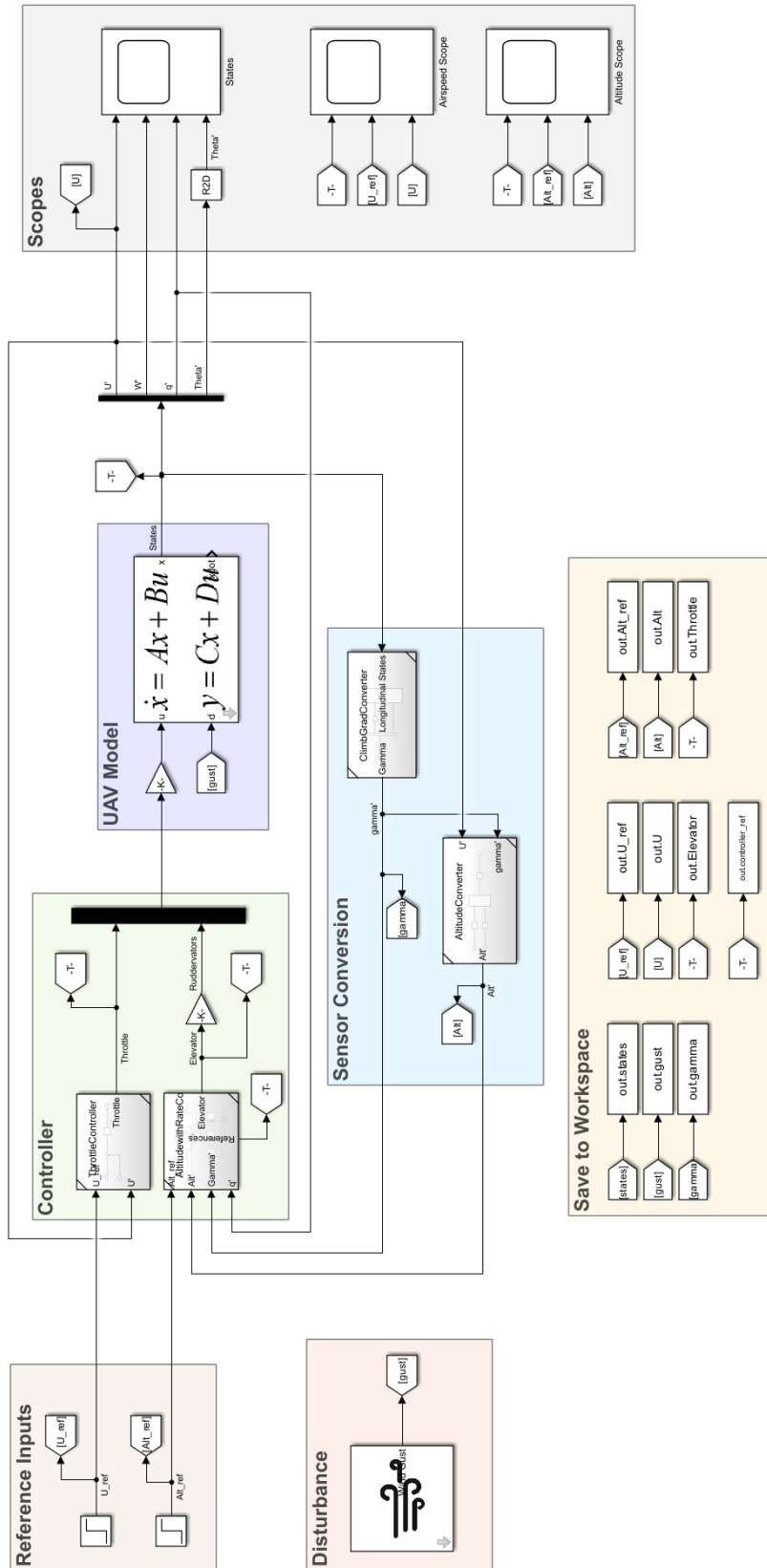


Figure 23: Simulink framework for verifying longitudinal controllers

E Flight Simulator Integrated Controller Verification

The integrated longitudinal controller, along with the lateral controller developed by Wilson [23], was implemented into the aircraft developed in the X-Plane 10 flight simulator developed by Chotai and Wales [10] [24]. Figures 24 and 25 show the step tracking performance of the controller which closely matched the Simulink predicted model in terms of open loop behaviour. Thrust was however significantly higher and thus, the climb gradient limit was increased to 10° . Figures 26 and 27 show the tracking performance of the controller when flown in gusty conditions of magnitude up to 15.2 m/s. Whilst altitude tracking remained largely stable and smooth, velocity tracking was quite poor. However, this was likely due to the nature of an aircraft's dynamic motion. Gust directly affects the airspeed; whereas, the altitude was not directly affected by gust and was compensated by a high frequency pitch rate controller on the most inner control loop.

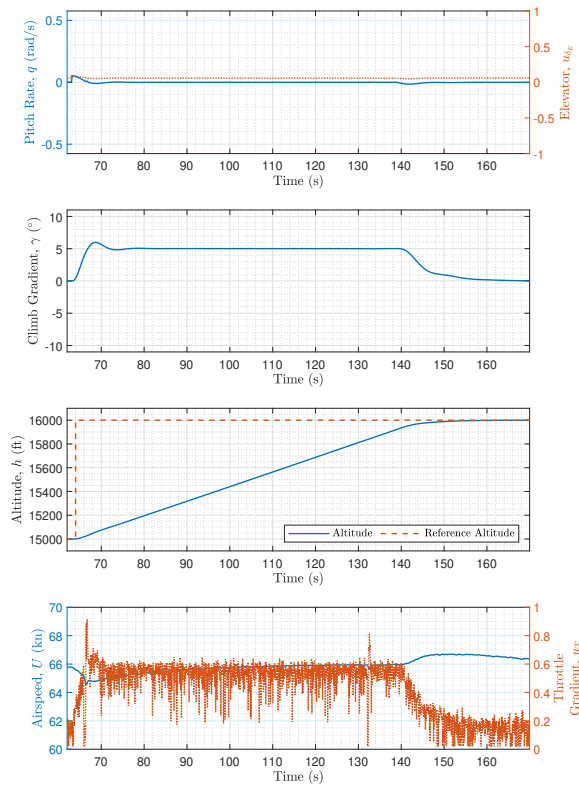


Figure 24: Step tracking of 1000ft Climb in the flight simulator in calm winds

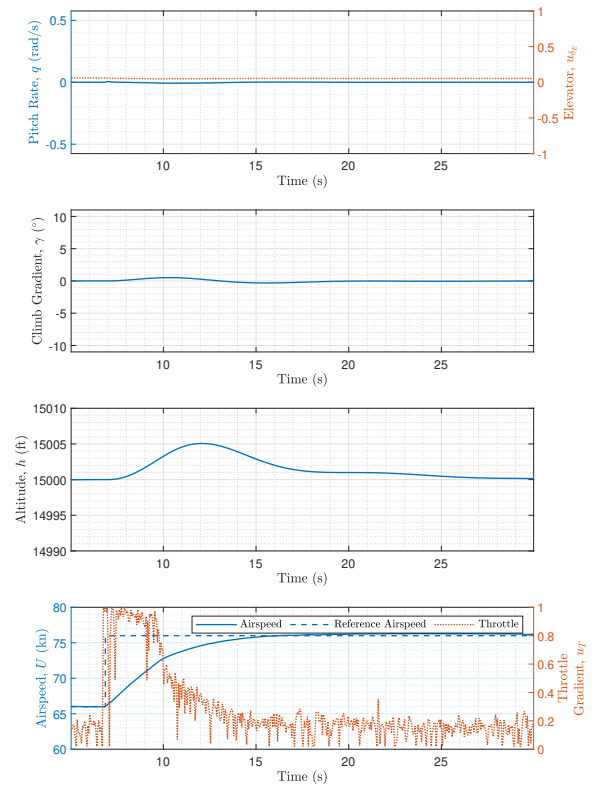


Figure 25: Step tracking of 10 kn speed increase in the flight simulator in calm winds

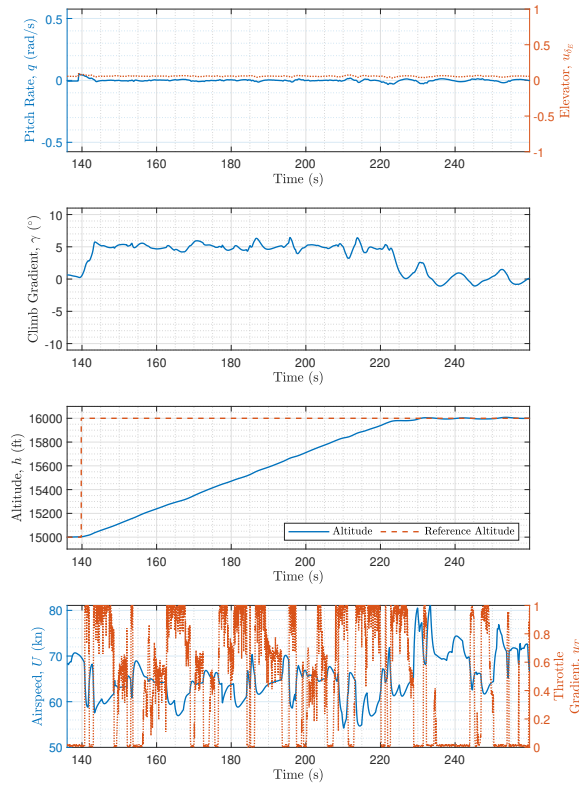


Figure 26: Step tracking of 1000ft Climb in the flight simulator in gust

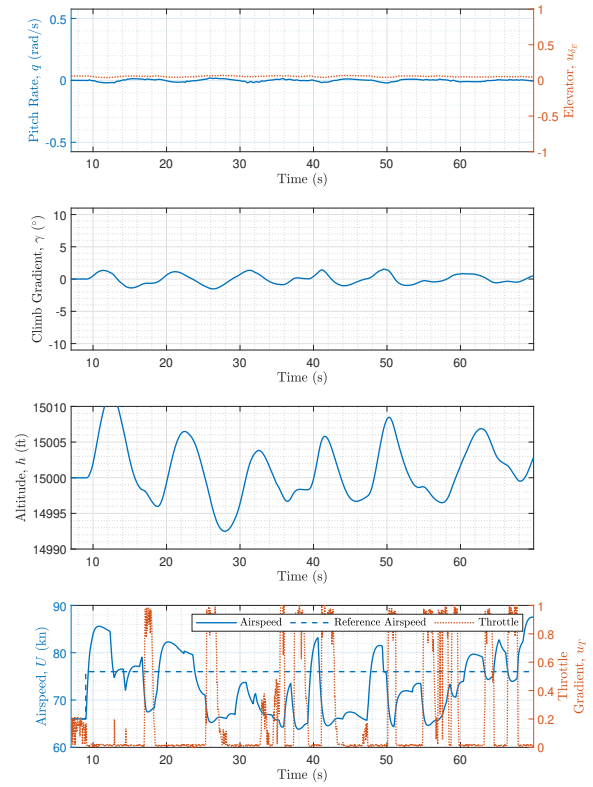


Figure 27: Step tracking of 10 kn speed increase in the flight simulator in gust